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Data Mining I

Semi-supervised classification

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AGENDA

- Cluster analysis
 - Semi-supervised classification
 - Nearest Neighbors
 - Decision Trees

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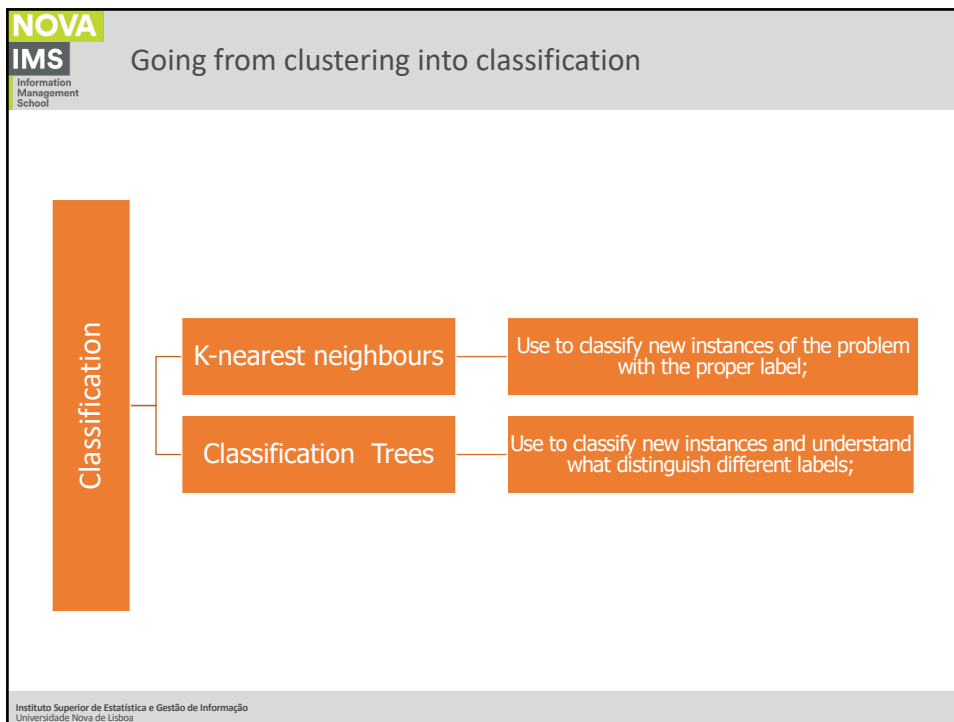
Going from clustering into classification

```
graph LR; A[Unlabeled Data] --> B[Clustering - Labels]; B --> C[Classification: KNN and Trees];
```

The diagram illustrates a three-step process for moving from clustering to classification. It features a large, light blue arrow pointing from left to right. Inside this arrow, there are three rounded rectangular boxes, each containing a step in the process:

- Unlabeled Data**: The starting point of the process.
- Clustering - Labels**: The intermediate step where data is grouped and labeled.
- Classification: KNN and Trees**: The final step, where specific classification algorithms are applied.

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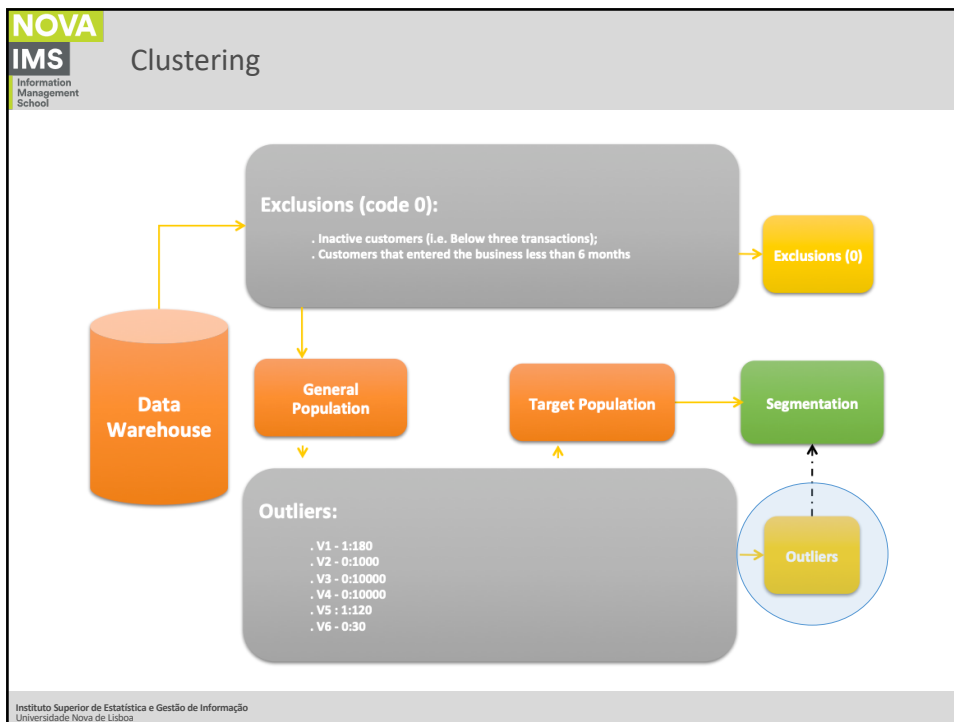
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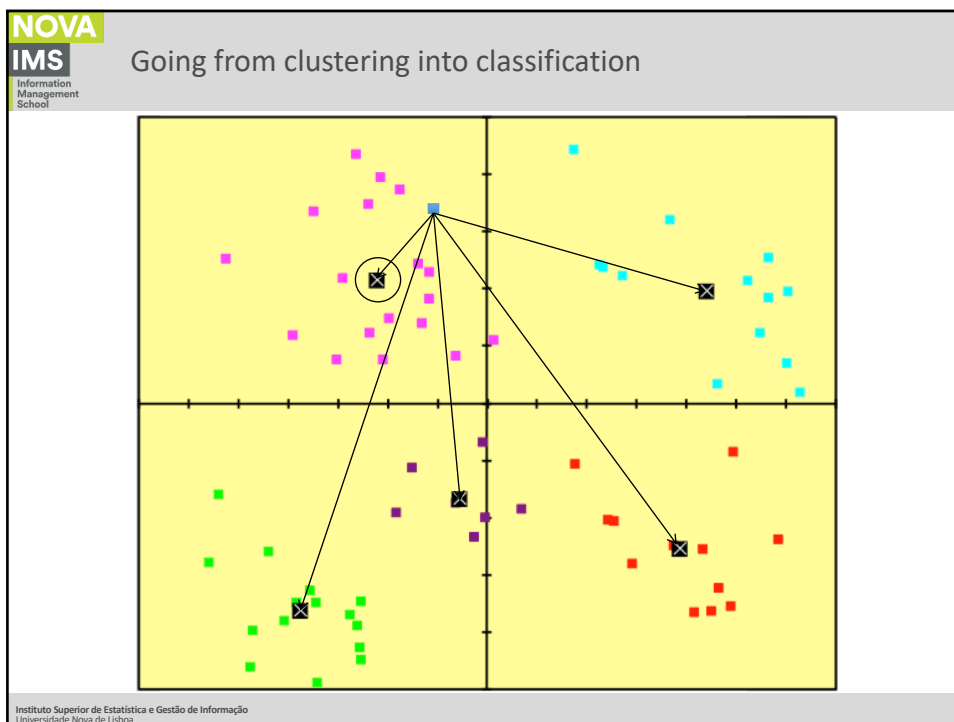
k-nearest neighbors (**k**-NN)

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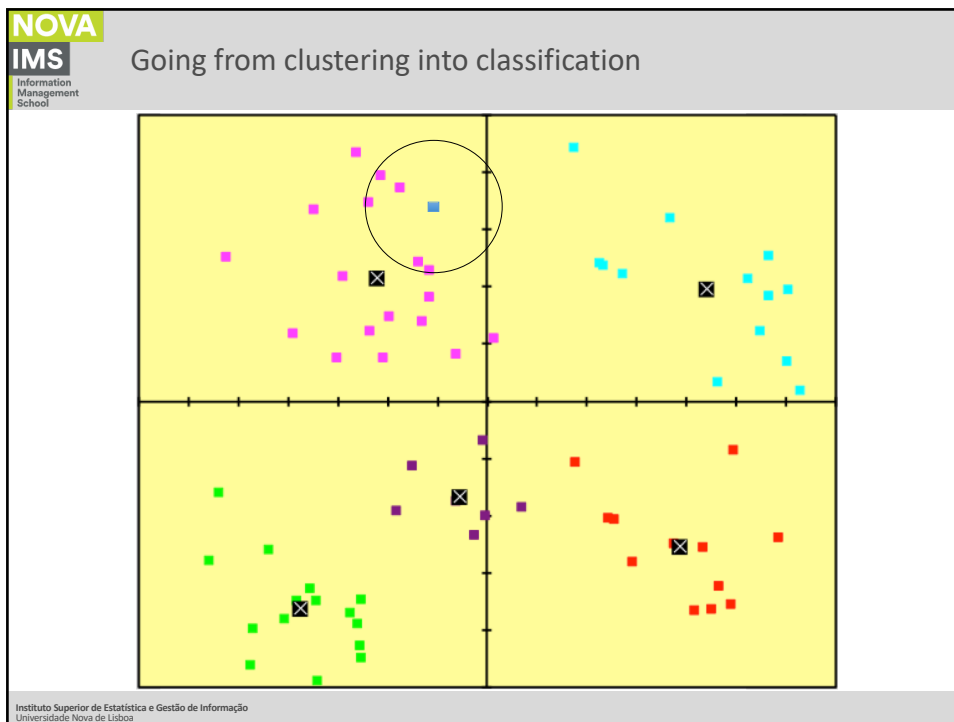
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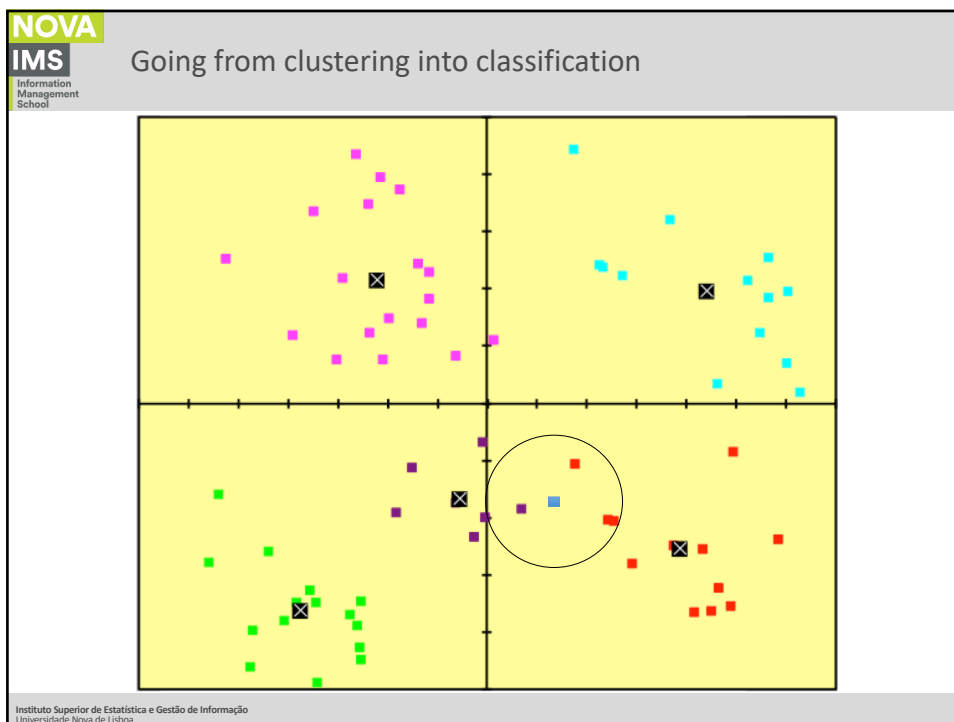
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k -nearest neighbors

- Instance based classification:
 - Simplest form of learning;
 - Training instances are searched for instances that most closely resembles new instance;
 - The instances themselves represent the knowledge;
 - Also called instance-based learning
 - Similarity function defines what's "learned"

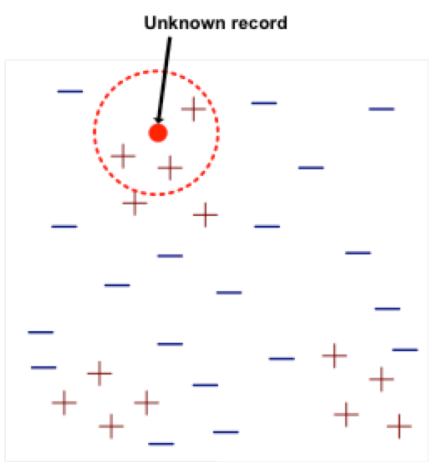
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k -nearest neighbors

- Requires three things:
 1. The set of stored records
 2. Distance Metric to compute distance between records
 3. The value of k , the number of nearest neighbors to retrieve



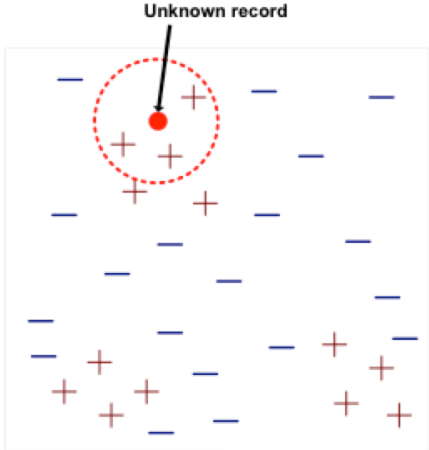
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k -nearest neighbors

- To classify an unknown record:
 1. Compute distance to other training records
 2. Identify k nearest neighbors
 3. Use class labels of nearest neighbors to determine the class label of unknown record (e.g., by taking majority vote)



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k -nearest neighbors

- Compute distance between two points:
 - Euclidean distance

$$d(p, q) = \sqrt{\sum_i (p_i - q_i)^2}$$
- Determine the class from nearest neighbor list
 - Take the majority vote of class labels among the k -nearest neighbors
 - Weigh the vote according to distance
 - weight factor, $w = 1/d^2$

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k -nearest neighbors

- k -nn frontiers (and the number k):
 - Large k
 - Smooth frontiers
 - Unable to detect small variations
 - Small k
 - Very sensitive to outliers
 - Crisp frontiers

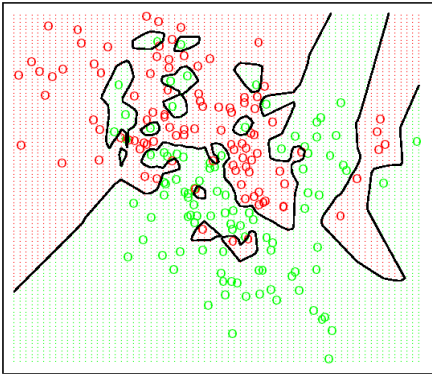
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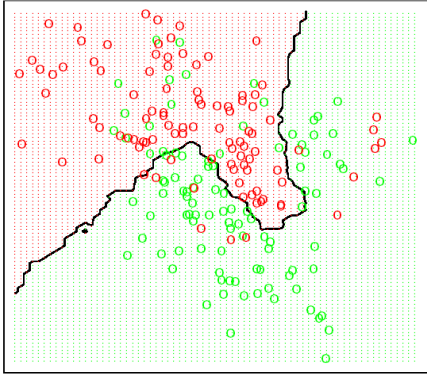
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k -nearest neighbors

1-nn



15-nn



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Classification Trees

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Classification Trees

- **Classification Trees:**
 - Classification tree are typically considered to be classification and regression tools
 - On of its most important advantages relates with the simplicity of the interpretation of its results
 - Thus, the end result of a classification tree can easily be expressed in English or SQL.

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Classification Trees

- A classification tree is a decisional algorithm
- It can be seen as a way of storing knowledge
- The objective is to discriminate between classes
- Obtain leaves as pure as possible
- If possible each leaf should represent only individuals from a specific class

Nodes (tests) → **Edges (results)** → **Leaves (terminal nodes)**

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Classification Trees

$$aprovado \Leftrightarrow (exame \geq 10)$$

$$aprovado \Leftrightarrow (exame < 10) \wedge (projecto \geq 14) \wedge (exame \geq 8)$$

$$reprovado \Leftrightarrow (exame < 10) \wedge (projecto < 14)$$

$$reprovado \Leftrightarrow (exame < 10) \wedge (projecto > 14) \wedge (exame < 8)$$

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Classification Trees

- **Classification Trees (strenghts):**
 - Interpretation
 - We can easily understand the reasons behind a specific classification decision
 - May use different types of data
 - Interval, ordinal, nominal, etc.
 - Insensitive to scale factors
 - Variables measured in different scales may be used without any type of normalization

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Classification Trees

- **Classification Trees (strenghts):**
 - Automatically defines the most relevant variables
 - These are the variables used at the top of the tree
 - Can be adapted to a regression
 - Each leave becomes a linear model

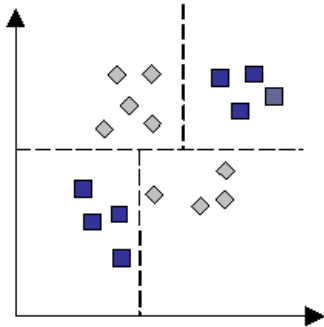
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Classification Trees

- **Classification Trees (strenghts):**
 - Boundaries are linear and perpendicular to the variables axys
 - Sensitive to small perturbations in the data



From Gahegan and West
http://divcom.otago.ac.nz/SIRC/GeoComp/GeoComp98/61/gc_61.htm

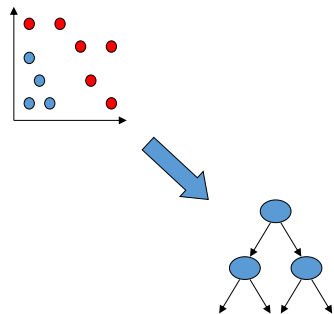
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Classification Trees

- **Classification Trees (strenghts):**
 - Build (induce) a tree from data
 - Problems:
 - What to do?
 - Which variable to use?
 - What partition to use?
 - Which node to split?
 - How many edges per node?
 - When to stop?



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Classification Trees

- **Classification Trees:**
 - ID3, C4.5 e C5 [Quinlan 86,93]
 - Iterative Dichotomizer 3
 - CART
 - Classification and regression trees [Breiman 84]
 - CHAID [Hartigan 75]
 - Used in SPSS and SAS...
 - Muitas (mesmo muitas) outras variantes...
 - In SAS you can choose different parameters to build your tree.

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Classification Trees

- **Classification Trees:**
 - In each level it divides the set into alternative partitions.
 - Using a measure of quality selects the best partition.
 - The process is repeated for each element of the partition.
 - Stops when a given criteria is reached

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Classification Trees

- **Classification Trees:**
 - It assumes the existence of a target variable “Class” meaning the examples were previously classified.
 - Each node specifies a unique attribute which is used as test.
 - N – node N
 - ASET – Attribute Set
 - ISET – Instance Set

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Classification Trees

If Se the ISET is empty then the terminal node N is an unknown class
 if not
 If all the examples of ISET are of the same class
 then the terminal node N has the name of the class
 if not
 For each attribute A of the set of attribute ASET
 Evaluate A according to its capability to discriminate a class
 Select the attribute B which has the best discriminate value
 For each value V of the best attribute B
 Create a new node C from node N
 Place the par attribute value (B, V) in C
 Let JSET be the set of examples of ISET with value V in B
 Let KSET be the set of attributes of ASET with B removed
 DDT(C, KSET, JSET)

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Classification Trees

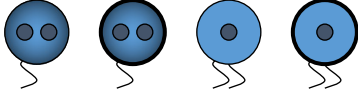
Lethargia



Burpoma



Healthy



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Classification Trees

Worked-Example General Idea

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Classification Trees

Table

# Núcleos	# Caudas	Cor	Membrana	Classe
1	1	Clara	Fina	<i>Lethargia</i>
2	1	Clara	Fina	<i>Lethargia</i>
1	1	Clara	Grossa	<i>Lethargia</i>
1	1	Escura	Fina	<i>Lethargia</i>
1	1	Escura	Grossa	<i>Lethargia</i>
2	2	Clara	Fina	<i>Burpoma</i>
2	2	Escura	Fina	<i>Burpoma</i>
2	2	Escura	Grossa	<i>Burpoma</i>
2	1	Escura	Fina	<i>Saudável</i>
2	1	Escura	Grossa	<i>Saudável</i>
1	2	Clara	Fina	<i>Saudável</i>
1	2	Clara	Grossa	<i>Saudável</i>

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Classification Trees

- **Classification Trees:**
 - Measure to discriminate the attribute

$$f(A) = \frac{1}{n} \sum_{i=1}^{|A|} C_i$$

- n is the total number of examples and C_i the number of examples correctly classified based on the most frequent class.
- This is a measure of “dominance” or “purity”

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Classification Trees

Table

# núcleos	1	2
Lethargia	4	1
Burpoma	0	3
Saudável	2	2

# Núcleos	# Caudas	Cor	Membrana	Classe
1	1	Clara	Fina	Lethargia
2	1	Clara	Fina	Lethargia
1	1	Clara	Grossa	Lethargia
1	1	Escura	Fina	Lethargia
1	1	Escura	Grossa	Lethargia
2	2	Clara	Fina	Burpoma
2	2	Escura	Fina	Burpoma
2	2	Escura	Grossa	Burpoma
2	1	Escura	Fina	Saudável
2	1	Escura	Grossa	Saudável
1	2	Clara	Fina	Saudável
1	2	Clara	Grossa	Saudável

Discrimination:

$$(4 + 3) / 12 = 0.58$$

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Classification Trees

Table

# caudas	1	2
Lethargia	5	0
Burpoma	0	3
Saudável	2	2

# Núcleos	# Caudas	Cor	Membrana	Classe
1	1	Clara	Fina	Lethargia
2	1	Clara	Fina	Lethargia
1	1	Clara	Grossa	Lethargia
1	1	Escura	Fina	Lethargia
1	1	Escura	Grossa	Lethargia
2	2	Clara	Fina	Burpoma
2	2	Escura	Fina	Burpoma
2	2	Escura	Grossa	Burpoma
2	1	Escura	Fina	Saudável
2	1	Escura	Grossa	Saudável
1	2	Clara	Fina	Saudável
1	2	Clara	Grossa	Saudável

Discrimination:

$$(5 + 3) / 12 = 0.67$$

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Classification Trees

Table

	Cor	Clara	Escura
Lethargia		3	2
Burpoma		1	2
Saudável		2	2

Discrimination:

$$(3 + 2) / 12 = 0.41$$

# Núcleos	# Caudas	Cor	Membrana	Classe
1	1	Clara	Fina	Lethargia
2	1	Clara	Fina	Lethargia
1	1	Clara	Grossa	Lethargia
1	1	Escura	Fina	Lethargia
1	1	Escura	Grossa	Lethargia
2	2	Clara	Fina	Burpoma
2	2	Escura	Fina	Burpoma
2	2	Escura	Grossa	Burpoma
2	1	Escura	Fina	Saudável
2	1	Escura	Grossa	Saudável
1	2	Clara	Fina	Saudável
1	2	Clara	Grossa	Saudável

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Classification Trees

Table

	Membrana	Fina	Grossa
Lethargia		3	2
Burpoma		2	1
Saudável		3	1

Discrimination:

$$(3 + 2) / 12 = 0.41$$

# Núcleos	# Caudas	Cor	Membrana	Classe
1	1	Clara	Fina	Lethargia
2	1	Clara	Fina	Lethargia
1	1	Clara	Grossa	Lethargia
1	1	Escura	Fina	Lethargia
1	1	Escura	Grossa	Lethargia
2	2	Clara	Fina	Burpoma
2	2	Escura	Fina	Burpoma
2	2	Escura	Grossa	Burpoma
2	1	Escura	Fina	Saudável
2	1	Escura	Grossa	Saudável
1	2	Clara	Fina	Saudável
1	2	Clara	Grossa	Saudável

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Classification Trees

Choice: # Caudas

# núcleos	1	2
Lethargia	4	1
Burpoma	0	3
Saudável	2	2

0.58

# caudas	1	2
Lethargia	5	0
Burpoma	0	3
Saudável	2	2

0.67

Cor	Clara	Escura
Lethargia	3	2
Burpoma	1	2
Saudável	2	2

0.41

Membrana	Fina	Grossa
Lethargia	3	2
Burpoma	2	1
Saudável	3	1

0.41

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Classification Trees

Initial Partition

Caudas

one

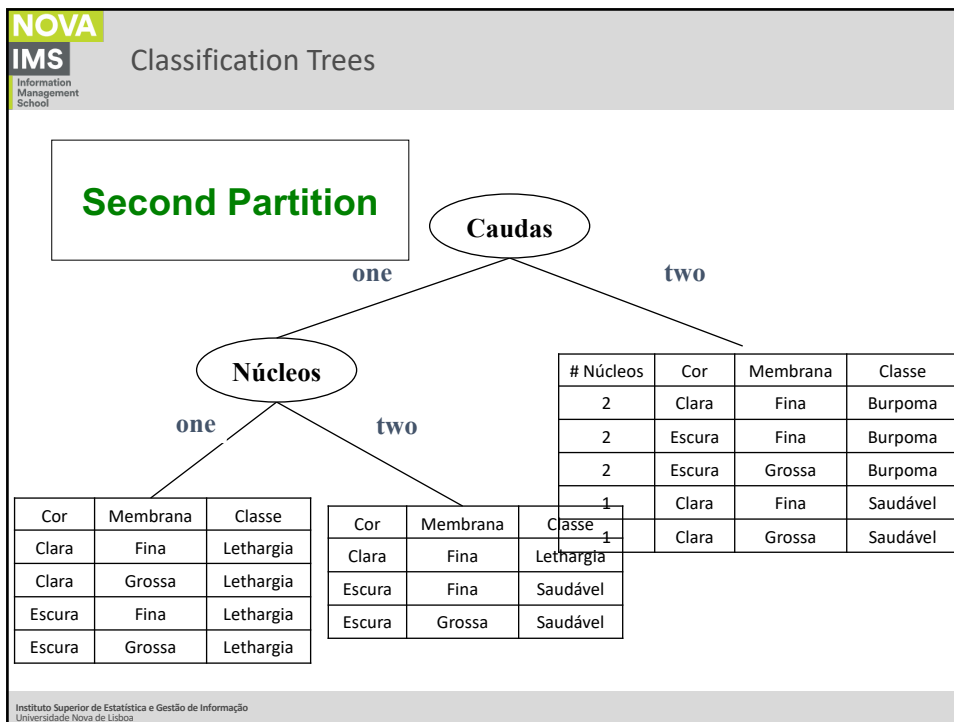
# Núcleos	Cor	Membrana	Classe
1	Clara	Fina	Lethargia
2	Clara	Fina	Lethargia
1	Clara	Grossa	Lethargia
1	Escura	Fina	Lethargia
1	Escura	Grossa	Lethargia
2	Escura	Fina	Saudável
2	Escura	Grossa	Saudável

two

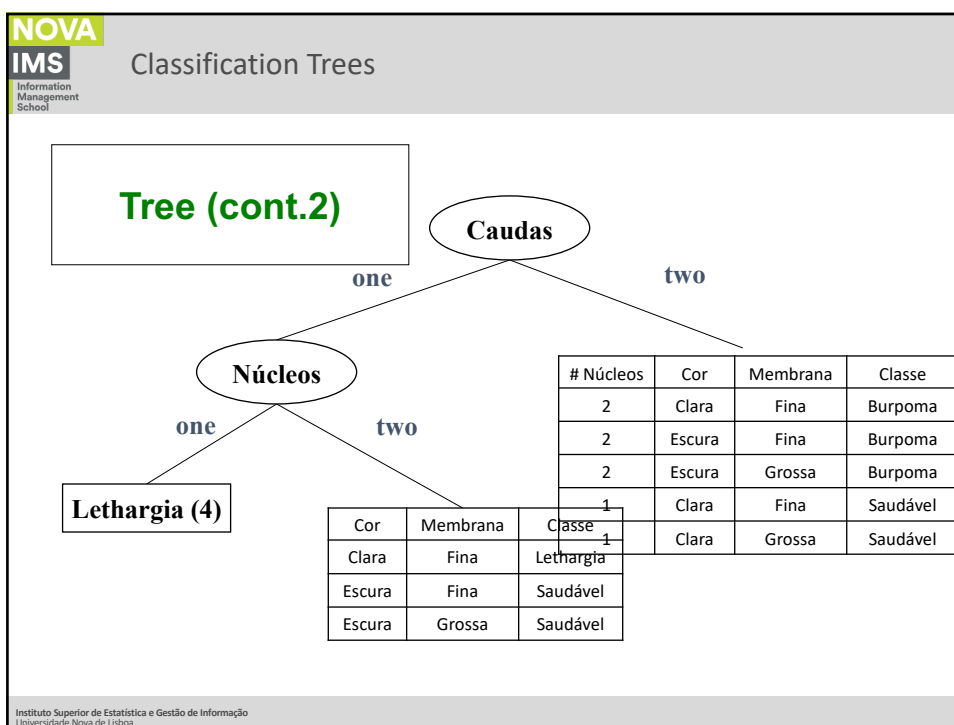
# Núcleos	Cor	Membrana	Classe
2	Clara	Fina	Burpoma
2	Escura	Fina	Burpoma
2	Escura	Grossa	Burpoma
1	Clara	Fina	Saudável
1	Clara	Grossa	Saudável

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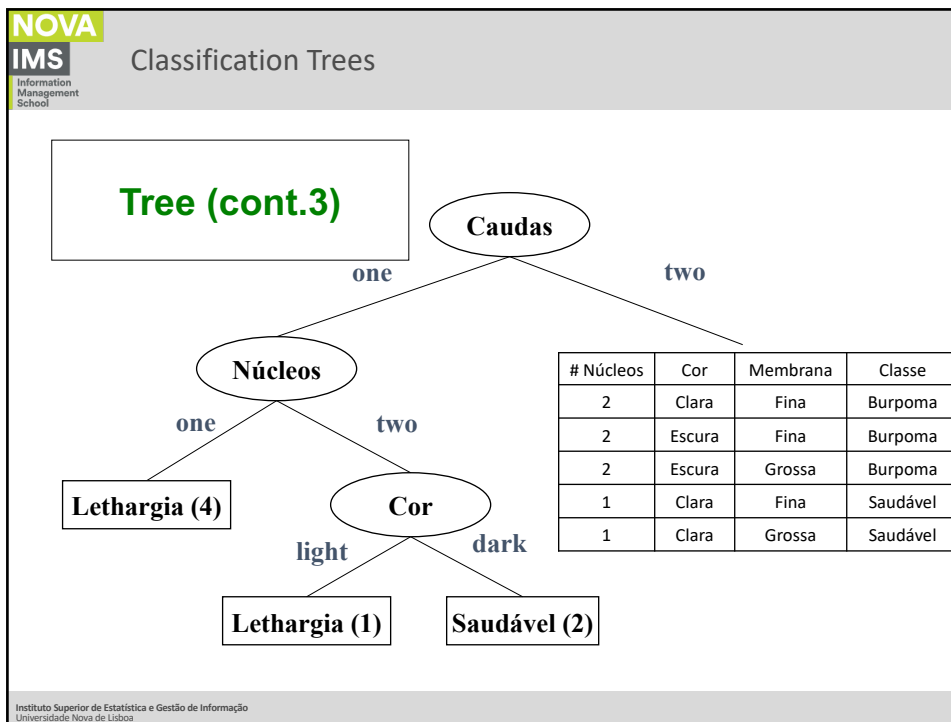
38



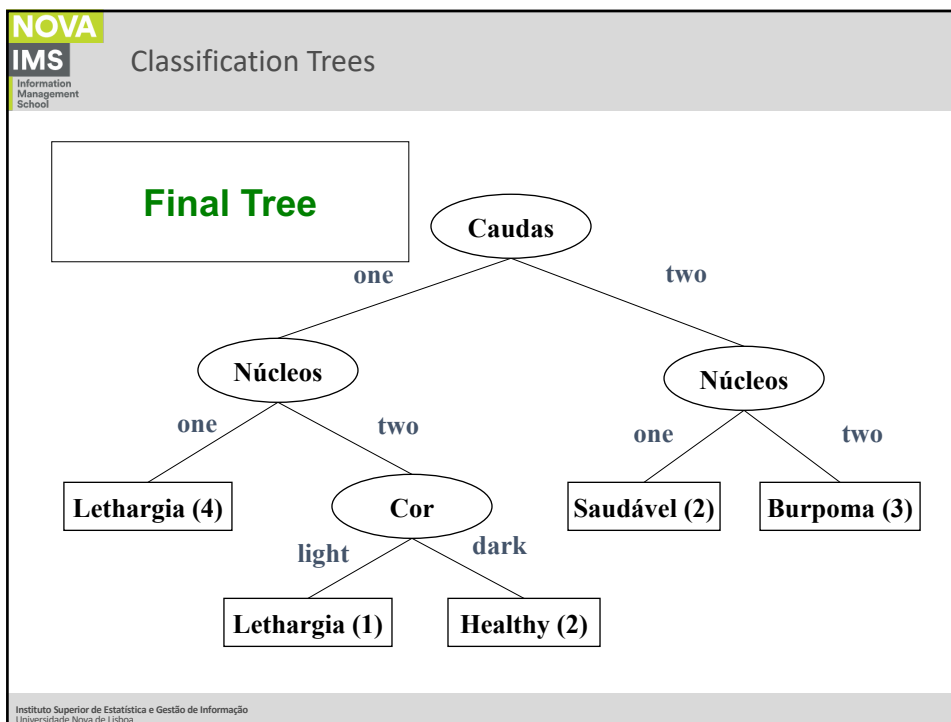
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Classification Trees

The description of the three classes

$$\begin{aligned} & (caudas = one) \wedge (nucleos = one) \\ & \vee \\ & (caudas = one) \wedge (nucleos = two) \wedge (cor = clara) \rightarrow Lethagia \\ & (caudas = duas) \wedge (nucleos = um) \\ & \vee \\ & (caudas = uma) \wedge (nucleos = 2) \wedge (cor = escura) \rightarrow Saudável \\ & (caudas = duas) \wedge (nucleos = dois) \rightarrow Burpoma \end{aligned}$$

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Classification Trees

**Worked-Example
Tree Accuracy**

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Classification Trees

Quality of the results

Error rate

100 100 100
A B C

100, 100, 100

75, 65, 20

25, 35, 80

7, 63, 4

68, 2, 16

10, 15, 70

15, 20, 10

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Classification Trees

Quality of the results

Error rate

100 100 100
A B C

100, 100, 100

75, 65, 20

25, 35, 80

7, 63, 4

68, 2, 16

10, 15, 70

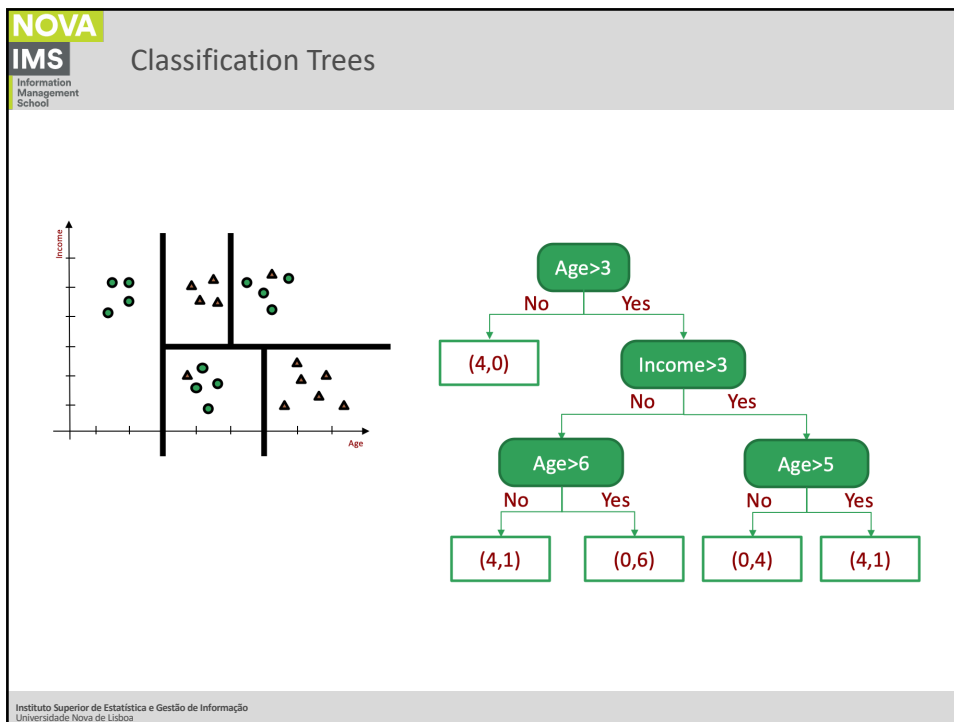
15, 20, 10

Error Calculation

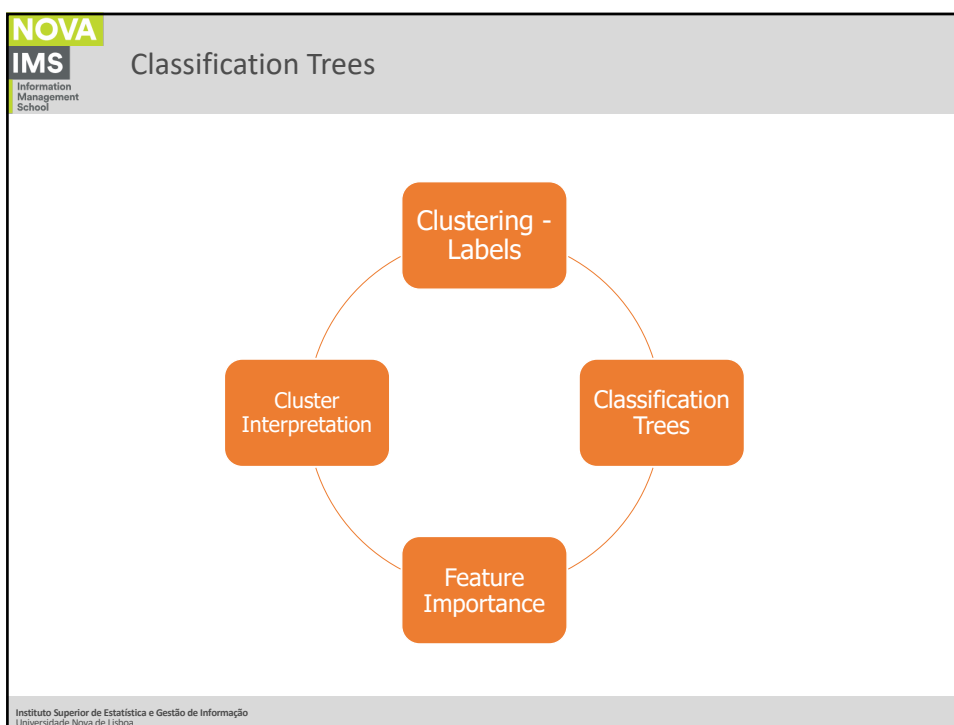
$$TEA = \frac{(74 * 14,9\% + 86 * 20,9\% + 95 * 26,3\% + 45 * 55,5\%)}{300} = 26,3\%$$

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Classification Trees

- **Classification Trees in Clustering:**
 - Feature Importance:
 - Train a classification tree on the original dataset, using the cluster assignments as the target variable.
 - The classification tree can help identify which features are important in distinguishing between the clusters. Look at the top nodes of the tree to identify the most critical features for distinguishing between clusters.
 - Interpretation:
 - Analyze the classification tree to understand how the features contribute to the formation of different clusters. This can provide insights into the characteristics that define each cluster.

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Classification Trees

- **Classification Trees in Clustering:**

```

graph TD
    Root["X[2] <= 0.311  
gini = 0.75  
samples = 4000  
value = [1000, 1000, 1000, 1000]"]
    Left["X[7] <= -3.849  
gini = 0.5  
samples = 2000  
value = [0, 0, 1000, 1000]"]
    Right["X[7] <= 3.841  
gini = 0.5  
samples = 2000  
value = [1000, 1000, 0, 0]"]
    Purple["gini = 0.0  
samples = 1000  
value = [0, 0, 0, 1000]"]
    Blue["gini = 0.0  
samples = 1000  
value = [0, 0, 1000, 0]"]
    Green["gini = 0.0  
samples = 1000  
value = [0, 1000, 0, 0]"]
    Orange["gini = 0.0  
samples = 1000  
value = [1000, 0, 0, 0]"]

    Root --> Left
    Root --> Right
    Left --> Purple
    Left --> Blue
    Right --> Green
    Right --> Orange
  
```

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Classification Trees

- **Classification Trees in Clustering:**
 - Visualizations:
 - Create visualizations to better understand the relationships between clusters and features.
 - Validate and Refine:
 - Use the insights gained from the classification tree to validate and refine your clustering solution.
 - Consider whether the identified features align with your domain knowledge and whether they make sense in the context of the problem you are trying to solve.

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Classification Trees

Worked-Example

Using Entropy to Decide on the Partitions

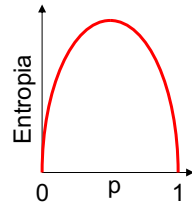
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Classification Trees

- **Classification Trees:**
 - Entropy
 - Measures the purity of a node through entropy, defined as:
$$E = -p \log_2(p)$$
 - Where p is the proportion of examples of a particular class



The graph shows Entropy on the y-axis and p on the x-axis. The x-axis ranges from 0 to 1, with a point p marked. A red curve starts at (0,0), rises to a peak at p=0.5, and then falls back to (1,0).

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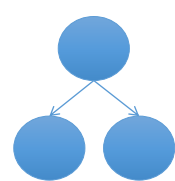
Classification Trees

- **Entropy for Classification Trees**
 - Entropy
 - **Partition entropy:**

$$Ent(S) = - \sum_{i=1}^{\#C} p_i \log_2(p_i)$$
 - **Gain in choosing A attribute:**

$$Gain(Ent_{new}) = Ent_{initial} - Ent_{new}$$

$$Gain(S, A) = Ent(S) - \sum_{v \in \text{Valores}(A)} \frac{\#S_v}{\#S} Ent(S_v)$$



The diagram shows a root node (blue circle) with two arrows pointing down to two child nodes (blue circles).

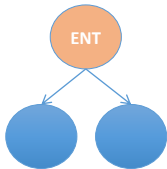
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Classification Trees

- Entropy for Classification Trees**
 - Entropy



	Income <= 50K	Income > 50K
	13	7

$$Ent(S) = -\left(\frac{13}{20} \log_2\left(\frac{13}{20}\right) + \frac{7}{20} \log_2\left(\frac{7}{20}\right)\right) = 0.403 + 0.530 = 0.934$$

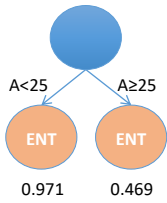
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	Income <= 50K	Income > 50K
Age < 25	4	6

$$Ent(Age < 25) = -\left(\frac{4}{10} \log_2\left(\frac{4}{10}\right) + \frac{6}{10} \log_2\left(\frac{6}{10}\right)\right) = 0.529 + 0.442 = 0.971$$

	Income <= 50K	Income > 50K
Age ≥ 25	9	1

$$Ent(Age \geq 25) = -\left(\frac{9}{10} \log_2\left(\frac{9}{10}\right) + \frac{1}{10} \log_2\left(\frac{1}{10}\right)\right) = 0.137 + 0.332 = 0.469$$

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Classification Trees

- Entropy for Classification Trees**
 - Entropy

	Income ≤ 50K	Income > 50K	#
Age < 25	9	1	10
Age ≥ 25	4	6	10

$$\sum_{v \in \text{Valores}(A)} \frac{\#S_v}{\#S} \text{Ent}(S_v) = \frac{1}{2}(0.469) + \frac{1}{2}(0.971) = 0.72$$

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Classification Trees

- Entropy for Classification Trees**
 - Entropy

	Income ≤ 50K	Income > 50K
Age < 25	9	1
Age ≥ 25	4	6

$$\text{Ent}(S) = -\left(\frac{13}{20} \log_2\left(\frac{13}{20}\right) + \frac{7}{20} \log_2\left(\frac{7}{20}\right)\right) = 0.403 + 0.530 = 0.934$$

$$\sum_{v \in \text{Valores}(A)} \frac{\#S_v}{\#S} \text{Ent}(S_v) = \frac{1}{2}(0.469) + \frac{1}{2}(0.971) = 0.72$$

$$\text{Gain}(S, A) = \text{Ent}(S) - \sum_{v \in \text{Valores}(A)} \frac{\#S_v}{\#S} \text{Ent}(S_v) = 0.934 - 0.72 = 0.214$$

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